



Insurance Premium — Test Results

Automated Test Documentation for Model Governance

MKM Research Labs

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1 Test Results — Insurance Premium (MKM-IP-001)

Tests run on **29 March 2026 at 10:29:14**.

Table 1: Test Results Summary — Insurance Premium (MKM-IP-001)

Total Tests	36
Passed	36
Failed	0
Skipped	0
Duration	0.00s

Table 2: Detailed Test Results — Insurance Premium

Test Description	Acceptance Criteria	Result	Time
All factors applied	<code>abs(result - round(expected, 2)) < 0.01</code>	Pass	0.000s
Basic calculation	<code>result == 1250.0</code>	Pass	0.000s
Claims factor increases premium	<code>with_claims > base</code>	Pass	0.000s
Clamped to max	<code>result == MAX_PREMIUM</code>	Pass	0.000s
Clamped to min	<code>result == MIN_PREMIUM</code>	Pass	0.000s
Contents premium added	<code>with_contents == base + 500.0</code>	Pass	0.000s
High flood risk increases premium	<code>high_flood > low_flood</code>	Pass	0.000s
Returns float	<code>isinstance(result, float)</code>	Pass	0.000s
Security factor reduces premium	<code>with_security < without</code>	Pass	0.000s
100 years is heritage	<code>get_age_premium_band(1925, current_year=2025) == 'heritage'</code>	Pass	0.000s
10 years is modern	<code>get_age_premium_band(2015, current_year=2025) == 'modern'</code>	Pass	0.000s
30 years is established	<code>get_age_premium_band(1995, current_year=2025) == 'establi...</code>	Pass	0.000s
Default current year	<code>result in AGE_PREMIUM_FACTORS</code>	Pass	0.000s
Established	<code>get_age_premium_band(1960, current_year=2025) == 'establi...</code>	Pass	0.000s
Exactly 29 years modern	<code>get_age_premium_band(1996, current_year=2025) == 'modern'</code>	Pass	0.000s
Exactly 99 years established	<code>get_age_premium_band(1926, current_year=2025) == 'establi...</code>	Pass	0.000s
Exactly 9 years new build	<code>get_age_premium_band(2016, current_year=2025) == 'new.build'</code>	Pass	0.000s
Heritage old	<code>get_age_premium_band(1850, current_year=2025) == 'heritage'</code>	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
Modern	<code>get_age_premium_band(2000, current_year=2025) == 'modern'</code>	Pass	0.000s
New build	<code>get_age_premium_band(2020, current_year=2025) == 'new.build'</code>	Pass	0.000s
Exactly 100 third band	<code>(lo, hi) == AREA_PREMIUM_BANDS[2][1]</code>	Pass	0.000s
Exactly 50 second band	<code>(lo, hi) == AREA_PREMIUM_BANDS[1][1]</code>	Pass	0.000s
Just below 100 second band	<code>(lo, hi) == AREA_PREMIUM_BANDS[1][1]</code>	Pass	0.000s
Just below 50 first band	<code>(lo, hi) == AREA_PREMIUM_BANDS[0][1]</code>	Pass	0.000s
Large area fourth band	<code>(lo, hi) == AREA_PREMIUM_BANDS[3][1]</code>	Pass	0.000s
Medium area second band	<code>(lo, hi) == AREA_PREMIUM_BANDS[1][1]</code>	Pass	0.000s
Returns tuple	<code>isinstance(result, tuple); len(result) == 2</code>	Pass	0.000s
Small area first band	<code>(lo, hi) == AREA_PREMIUM_BANDS[0][1]</code>	Pass	0.000s
Age premium factors all positive	<code>lo > 0; hi >= lo</code>	Pass	0.000s
Area bands are sorted	<code>thresholds == sorted(thresholds)</code>	Pass	0.000s
Base rate range	<code>lo > 0; hi > lo</code>	Pass	0.000s
Construction type factors present	<code>'Brick' in CONSTRUCTION_TYPE_FACTORS; 'Timber Frame' in CONSTRUCTION_TYPE_FACTORS</code>	Pass	0.000s
Flood risk factors increase with risk	<code>very_high_hi > very_low_lo</code>	Pass	0.000s
Flood risk factors present	<code>set(FLOOD_RISK_PREMIUM_FACTORS.keys()) == expected</code>	Pass	0.000s
Min max premium bounds	<code>MIN_PREMIUM == 200; MAX_PREMIUM == 20_000</code>	Pass	0.000s
Property type factors present	<code>set(PROPERTY_TYPE_PREMIUM_FACTORS.keys()) == expected</code>	Pass	0.000s

1.1 Test Document History

Date	Version	Description	Author
29-Mar-2026	1.0	Initial test suite (36 tests)	D. Kelly